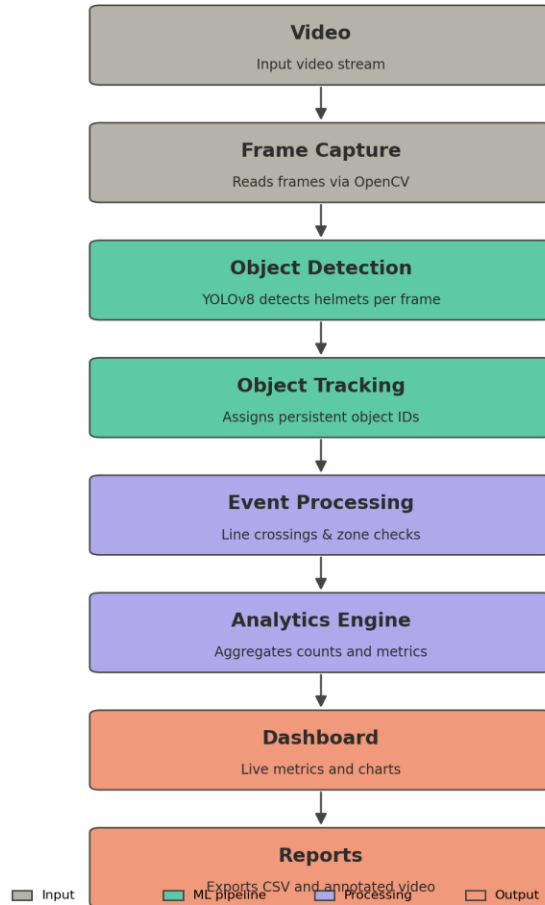


System Architecture

Helmet Detection & Tracking Analytics



Component Explanations

1. Video

The raw input source: an uploaded video file (or, in principle, a live camera or RTSP feed). This stage is simply the byte stream the system receives; no processing occurs here yet.

2. Frame Capture

OpenCV's VideoCapture reads the video one frame at a time in a processing loop. Each frame is handed off as a raw image array to the next stage. This is also where video resolution can be adjusted before further processing, as tested in the resolution experiments.

3. Object Detection

A custom-trained YOLOv8s model scans each frame and outputs bounding boxes with class labels (helmet / no_helmet) and confidence scores. This stage is stateless: it has no memory of previous frames and knows only what is present in the current image, not who or what it was a moment ago.

4. Object Tracking

This is where identity enters the pipeline. ByteTrack (or, alternatively, BoT-SORT) takes the current frame's detections and matches them against tracks carried over from previous frames, based on motion and, in BoT-SORT's case, appearance. A matched detection keeps its existing tracker ID; an unmatched detection may spawn a new ID, and a track that goes briefly unmatched is kept alive for a configurable number of frames before being dropped.

5. Event Processing

Once objects are identified and tracked, this stage checks for meaningful events: has a tracked object's position crossed a user-defined virtual counting line? Is it currently inside a defined region-of-interest zone? These checks run every frame and update running counters for entries, exits, and zone occupancy.

6. Analytics Engine

This layer accumulates per-frame results into overall statistics: total unique objects seen, currently active objects, average detection confidence, processing speed (FPS), and per-zone dwell time. It converts frame-by-frame detections into meaningful, whole-video analytics.

7. Dashboard

A live web interface (built with Streamlit) surfaces these analytics as they are computed: metric cards for active/total/entered/exited counts and FPS update in real time during processing, alongside a time-series chart of object counts and a summary chart once processing completes.

8. Reports

The final export stage produces an annotated video file (with all boxes, IDs, and movement trails drawn in) and a CSV report listing every tracked object's class, confidence, and duration tracked — the underlying data behind everything shown on the dashboard.